

Azure Data Factory

Questions and answers from many sources — de-duplicated and organised by section

Questions	4
Sections	4
AI-written answers	4 (marked  — the source had the question only)
Last updated	2026-09-30

Sources merged into this file

- `adf_open_questions.txt` — 4

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Fundamentals & Components (1)

Q1. What is the difference between a linked service and a dataset? 📌

A linked service is the connection (where): server, database, authentication, usually via Key Vault or managed identity. A dataset is the data (what): a specific table, file or folder, its format and schema, reached through a linked service. One linked service can serve many datasets.

Source: AI-written answer · question from adf_open_questions.txt

Metadata-driven Pipelines (1)

Q2. How would you design a metadata-driven pipeline that loads 100 tables without creating 100 pipelines? 📌

Use one generic pipeline driven by a control table instead of 100 pipelines.

- Control table: source schema/table, target path, load type (full/incremental), watermark column, active flag.
- Lookup reads the active rows → ForEach (parallel, batch count tuned to what the source can handle) → Copy activity with parameterised source and sink datasets (@item().table_name).
- A stored procedure updates the watermark and writes an audit row per table.

Adding a table is just a new row — no pipeline change or redeploy.

Source: AI-written answer · question from adf_open_questions.txt

Incremental Loads & Triggers (1)

Q3. How would you implement an incremental load using a watermark column? 📌

Store the last successful watermark (max modified_date) in a control table.

1. Lookup the old watermark and the new maximum from the source.
2. Copy only rows with modified_date > old AND <= new.
3. MERGE into the target on the business key.
4. Update the watermark only after the copy succeeds, so a failed run is simply retried.

Use a small overlap window plus MERGE to catch late commits without duplicates.

Source: AI-written answer · question from adf_open_questions.txt

Performance & Monitoring (1)

Q4. A daily file does not arrive on time. How would you alert instead of failing the pipeline? 📌

Don't let a missing file fail the pipeline.

- Validation activity (timeout + sleep interval), or Get Metadata with the 'exists' field inside an Until loop, to wait for the file.
- If Condition: when the file exists → run the copy; when it doesn't → call a Logic App/webhook that emails or posts to Teams 'file not received for <date>', log a MISSING status and end cleanly.
- Optionally switch to a storage event trigger so the pipeline starts when the file actually lands.

Source: AI-written answer · question from adf_open_questions.txt
